Screening procedure for selection of putative ligands of SARS-CoV-2 proteins

Consiglio Nazionale delle Ricerche

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INTRODUCTION

An acute respiratory disease, caused by a novel coronavirus (SARS-CoV-2) has spread throughout China and received worldwide attention. On 11 March 2020, World Health Organization (WHO) officially declared the COVID-19 epidemic as pandemic (*Fig. 1*). To date, there is no specific drug registered to combat this disease.

Our team is working on the screening of putative ligands of SARS-CoV-2 proteins whose structure is available (*Fig. 2*). Results of the screening highlighted a number of ligands with potential ability to bind the viral proteins, with a perspective interest in finding useful molecules against the pandemic urgency. In this poster we focus the attention on three compounds, currently used for the treatment of other pathologies, object of investigation during the first phase of the COVID-19 emergency, for their potential use against SARS-CoV-2.

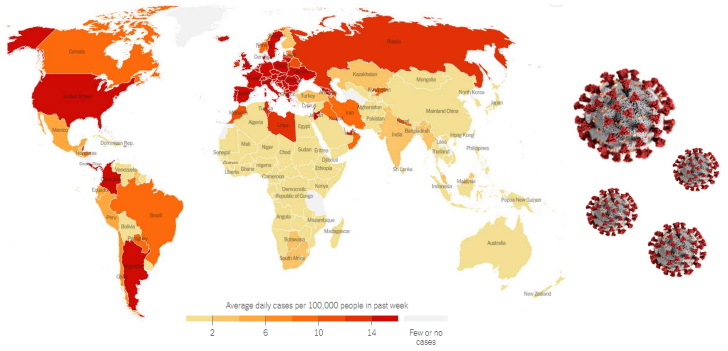


Fig.1: Hot spots map updated on 4/11/2020 related to the epidemiological situation of the COVID-19 pandemic by country.

Image sources: The Center for Systems Science and Engineering at Johns Hopkins University; National Health Commission of the People's Republic of China; World Health Organization. New York Times.

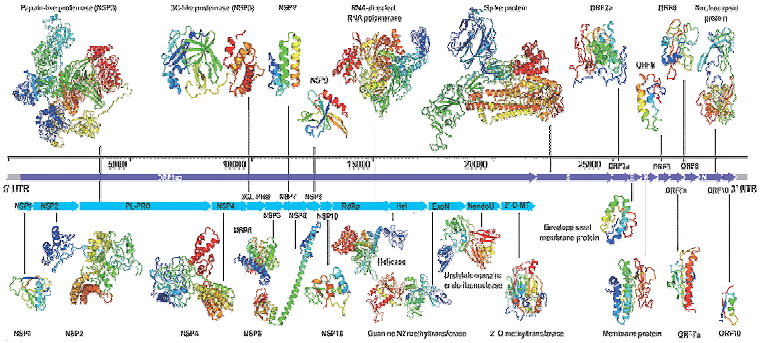


Fig. 2: Genome-wide structure and function modeling of SARS-CoV-2 proteins.

Image source: Zhang Lab https://zhanglab.ccmb.med.umich.edu/COVID-19/

METHODS

The investigation has been performed by means of a semi-automatic procedure (*Fig. 3*) that includes molecular docking simulations by AutoDock 4.2.6 (blind and focused) with a large number of ligands against the PDB structures available for proteins from the SARS-CoV-2. The semi-automatic procedure is configured to work on a Linux server and, depending on the server computational power, it may produce thousands of protein-ligand simulations in 24 hours.

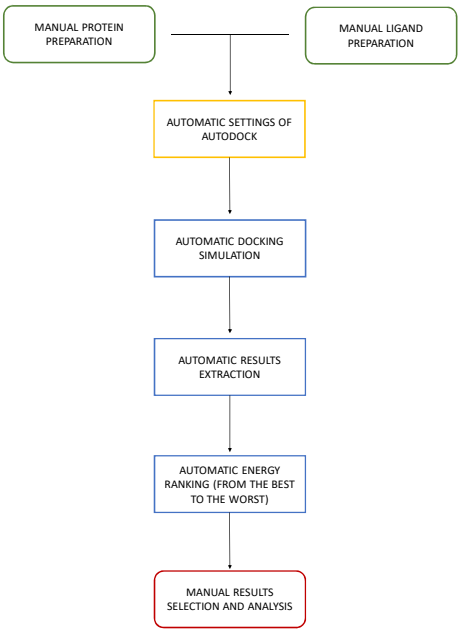


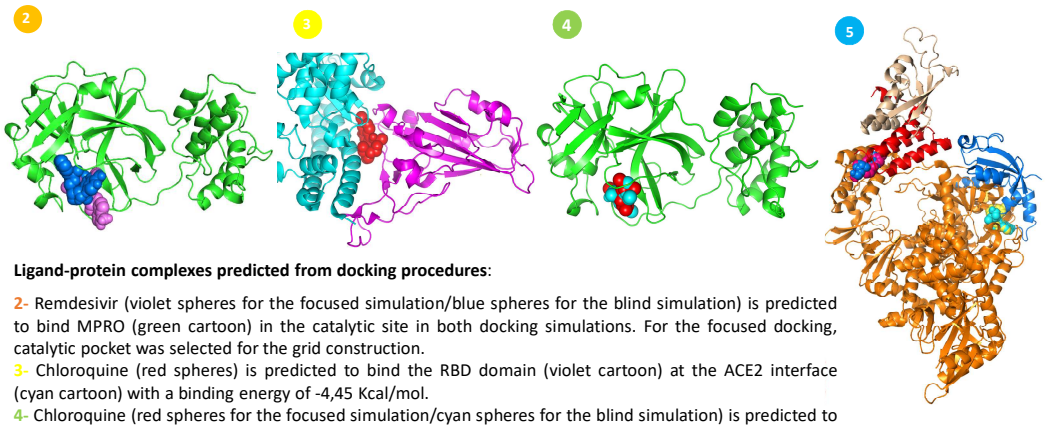
Fig. 3 : Workflow of the procedure applied.

RESULTS

1 Best docking results for the three drugs (remdesivir, chloroquine, hydroxychloroquine) towards the three fundamental protein of Sars-CoV-2: Spike (RBD domain: helps the virus to invade the cell), M-PRO (the main protease: makes most of the cuts that allow other NSP proteins to activate and promote viral life cycle and replication), NSP12 (viral RNA polymerase)(*Table 1*)

	REMDESIVIR				CHLOROQUINE				HYDROXYCHLOROQUINE			
	focused		blind		focused		blind		focused		blind	
PDB code	Mean Binding Energy (Kcal/mol)	Num. in cluster	Mean Binding Energy (Kcal/mol)	Num. in cluster	Mean Binding Energy (Kcal/mol)	Num. in cluster	Mean Binding Energy (Kcal/mol)	Num. in cluster	Mean Binding Energy (Kcal/mol)	Num. in cluster	Mean Binding Energy (Kcal/mol)	Num. in cluster
6m0j (SPIKE)	-	-	-4,68	1	-	-	-5,62	10	-	-	-4,54	12
			-3,9	2			-4,45	1			-4,31	1
6lu7 (MPRO)	-8,33	2	-3,99	1	-6,84	24	-6,04	8	-6,46	15	-5,52	4
7btf (NSP12)	-5,92	1	-3,46	1	-6,3	1	-4,45	8	-4,91	2	-4,5	2
	-	-	-	-	-5,24	4	-	-	-5,81	1	-4,16	2

Table 1: Molecular docking energy and related cluster's numerosity of selected putative inhibitors towards Sars-CoV-2 proteins.



Ligand-protein complexes predicted from docking procedures:

2- Remdesivir (violet spheres for the focused simulation/blue spheres for the blind simulation) is predicted to bind MPRO (green cartoon) in the catalytic site in both docking simulations. For the focused docking, catalytic pocket was selected for the grid construction.

3- Chloroquine (red spheres) is predicted to bind the RBD domain (violet cartoon) at the ACE2 interface (cyan cartoon) with a binding energy of -4,45 Kcal/mol.

4- Chloroquine (red spheres for the focused simulation/cyan spheres for the blind simulation) is predicted to bind MPRO (green cartoon) in the catalytic site in both docking simulations. For the focused docking, catalytic pocket was selected for the grid construction.

5- Hydroxychloroquine (blue or cyan spheres for the focused simulation/magenta or yellow spheres for the blind simulation) is predicted to bind NSP12 (orange cartoon) at NSP8 interface (red and blue cartoon).

CONCLUSION AND FUTURE WORKS

For the present work we have selected remdesivir, chloroquine and hydroxychloroquine, in order to verify their ability to interact with the main three viral proteins, due to the interest that these drugs raised up for their possible application in Sars-Cov-2 infection treatment. In agreement with the literature, remdesivir binds M-PRO in the catalytic site, with a favourable binding energy in the case of focused docking. However, although blind docking results identify the same binding pocket, they show a disagreement with the binding energy values, which are very unfavourable. Chloroquine, known to interact with human ACE2 protein, shows a weak interaction also with RBD domain of Spike protein. Anyway, the best target for this compound appears to be by M-PRO, for which both blind and focused docking are in good agreement. Docking simulations using hydroxychloroquine, even if detect correct and useful pockets, show high energy values that suggest that this molecule does not bind these proteins.

Our study is under further development to improve the semi-automatic procedure, as well as to analyze in details the results of docking simulations. Preliminary results suggest that other compounds, already considered useful against viral infections, could be more effective than remdesivir, chloroquine, and hydroxychloroquine, to target SARS-Cov2 proteins and interfere with molecular mechanisms related to the virus action.